

INTERNSHIP REPORT PRESENTATION

NeoLnk

An AI-Powered URL Shortener with Real-Time Analytics

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Agenda

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Introduction

This report documents the design, development, and verification of an AI-powered URL Shortener application named “NeoLnk,” built during a Python-focused internship conducted under The IoT Academy, Uniconverge Technologies, and upSkill Campus, in collaboration with AICTE.

The project involved building a complete web-based URL shortening service with a Python Flask backend, a SQLite database, and a modern, animated front-end dashboard for link management and analytics — covering custom aliases, expiration, QR codes, and real-time click tracking.

PROGRAM SNAPSHOT

29

Days of hands-on internship

9/9

Automated unit tests passed

3

Application layers built end-to-end

Organization Profile

1

AICTE

All India Council for Technical Education — apex statutory body overseeing technical education programs across India.

2

Uniconverge Technologies

IoT and technology solutions provider; delivers industry training through The IoT Academy to bridge academia and real-world skills.

3

upSkill Campus

Partners with AICTE to provide structured, project-based internship and skilling programs to engineering students nationwide.

Problem Statement & Objectives

THE PROBLEM

Long URLs are hard to share, remember, and track. Off-the-shelf shortening services often lack transparency, customization, and self-hosted control over data — organizations need a lightweight platform they can deploy and control independently, with full visibility into how links are used.

PROJECT OBJECTIVES

- ✓ Generate unique, collision-free short codes (Base62)
- ✓ Support custom aliases with collision handling
- ✓ Enforce link expiration constraints
- ✓ Track clicks with browser & platform detection
- ✓ Build a responsive, real-time analytics dashboard
- ✓ Validate via automated and manual testing

Technologies & Tools Used

Backend

Python

Flask

user-agents (UA parsing)

Database

SQLite

Base62 encoding logic

Frontend

HTML5 + Vanilla CSS

JavaScript

Outfit Google Font

Visualization & Testing

Chart.js

QRCode.js

Python unittest

System Architecture

Presentation layer

`index.html · main.js · styles.css`

↓ *REST API*

Application layer

`app.py` — Flask REST endpoints

↓ *DB queries*

Data layer

`database.py` · SQLite (links, click logs)

Three-tier architecture: a glassmorphism dashboard calls Flask REST endpoints, which read and write link and click data in SQLite, feeding analytics back to the dashboard in real time.

Development Methodology

1

Requirements & Setup

Configured requirements.txt with Flask and user-agents

2

Database Design

Built schema, Base62 short-code logic in database.py

3

Backend API

Implemented shorten, list, redirect & analytics endpoints

4

Frontend Build

Dashboard, QR rendering, Chart.js visualizations

5

Automated Testing

9 unit tests: aliasing, expiry, UA parsing, routing

6

Manual Verification

End-to-end browser testing of full user flow

Testing & Verification

AUTOMATED TESTS

9/9

tests passed

- Base62 code validation
- Custom alias collision handling
- Expiration constraints
- User-Agent browser/platform parsing
- REST API input & status codes
- Redirection behaviour

MANUAL BROWSER VERIFICATION

Generation	Shortened https://wikipedia.org to custom alias “wiki2”
QR & Copy	QR code displayed instantly; copy-to-clipboard toast confirmed
Redirection	Verified routing to Wikipedia; click count incremented to 1
Analytics	Click history, browser (Chrome) and OS charts rendered correctly

Results & Outcomes

The completed NeoLnk application meets all defined objectives: it reliably generates unique short links, supports custom aliases with collision handling, enforces expiration, and accurately tracks clicks with visitor browser and platform information — all validated through automated and manual testing.



Reliable link generation

Base62 short codes with alias collision handling



Expiration control

Links auto-expire based on configured constraints



Click analytics

Real-time browser & OS breakdown via Chart.js



Polished UX

Glassmorphism dashboard with instant QR generation

Internship Completion Certificate



Thank You

Questions & Discussion

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